



Constructing triangles

DRAWING (CONSTRUCTING) TRIANGLES REQUIRES A COMPASS, A RULER, AND A PROTRACTOR.

To construct a triangle, not all the measurements for its sides and angles are required, as long as some of the measurements are known in the right combination.

What is needed?

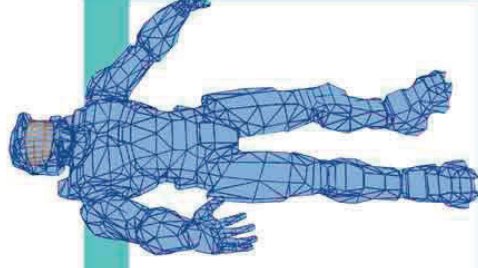
A triangle can be constructed from just a few of its measurements, using a combination of the tools mentioned above, and its unknown measurements can be found from the result. A triangle can be constructed when the measurements of all three sides (SSS) are known, when two angles and the side in between are known (ASA), or when two sides and the angle between them are known (SAS). In addition, knowing either the SSS, the ASA, or the SAS measurements of two triangles will reveal whether they are the same size (congruent)—if the measurements are equal, the triangles are congruent.

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Using triangles for 3-D graphics

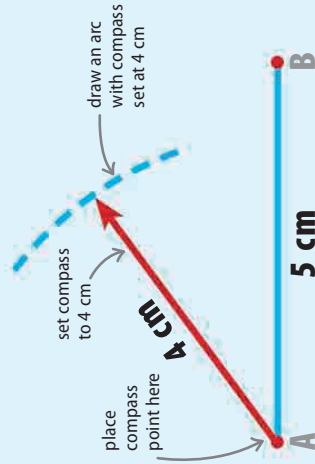
3-D graphics are common in films, computer games, and the internet. What may be surprising is that they are created using triangles. An object is drawn as a series of basic shapes, which are then divided into triangles. When the shape of the triangles is changed, the object appears to move. Each triangle is colored to bring the object to life.

Computer animation

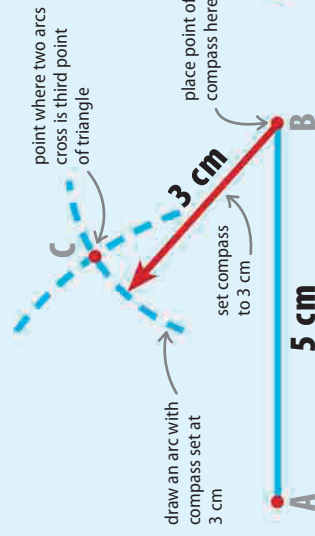
To create movement, a computer calculates the new shape of millions of shapes.

Constructing a triangle when three sides are known (SSS)

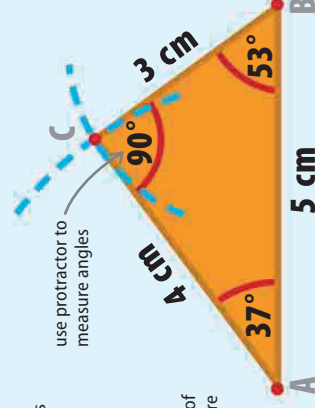
If the measurements of the three sides are given, for example, **5 cm**, **4 cm**, and **3 cm**, it is possible to construct a triangle using a ruler and a compass, following the steps below.



Draw the baseline, using the longest length. Label the ends A and B. Set the compass to the second length, 4 cm. Place the point of the compass on A and draw an arc.



Set the compass to the third length, 3 cm. Place the point of the compass on B and draw another arc. Mark the spot where the arcs intersect (cross) as point C.



Join the points to complete the triangle. Now use a protractor to find out the measurements of the angles. These will add up to 180° ($90^\circ + 53^\circ + 37^\circ = 180^\circ$).